DOCUMENT 00

10T SAFETY-CAPSULE DISTRIBUTED-LIFT AIRCRAFT

Conceptual Design Basis / Founder Edition

Founder	Amirbahador Atai
Program	10-ton class distributed-lift aircraft
Primary configuration	7 × 5.6 m rotors; fixed integrated safety capsule
Fuel basis	1,600 L nominal tank capacity
Document	00
Revision	0.1 — Conceptual Design Basis
Status	Preliminary / non-certified / not for flight or fabrication
Date	13 August 2026

DESIGN INTENT

A deliberate shift from the earlier 200-ton concept toward a smaller, buildable demonstrator. The architecture preserves the program’s central differentiator: a protected occupant capsule integrated into a multi-layer survival system rather than treating occupant protection as an afterthought.

IMPORTANT: This document is a system-level concept record. It is not an approved aircraft design, flight manual, structural substantiation package, engine installation approval, or certification basis. All numerical values are preliminary targets requiring analysis, testing, and qualified aerospace engineering review before hardware decisions.

1. Executive Design Statement

The program is now defined around a 10,000 kg maximum design mass target rather than the cancelled 200-ton architecture. The smaller aircraft is intended to be easier to prototype, demonstrate, discuss with investors and prospective industrial partners, and iterate without carrying the full complexity of the original heavy platform.

The configuration shown in the supplied project material is retained as the baseline: seven large rotors, approximately 5.6 m rotor diameter each, a compact 17 m × 17 m × 4 m overall envelope, a 1,600 L fuel tank, and a protected central occupant capsule.

Two capsule variants remain part of the broader family. Document 00 defines the simpler flagship demonstrator: a permanently integrated capsule with substantial mechanical energy-attenuation interfaces. The separable/ejectable capsule is treated as a future derivative and is not the primary architecture in this document.

Core principle: simplify the aircraft, not the safety philosophy. The vehicle becomes smaller and more buildable while the safety architecture remains deliberately layered.

2. Program Baseline and Configuration

The following values are treated as program inputs from the supplied material and the latest user-defined revision. They are not represented as certified performance claims.

Parameter	Baseline target	Status / note
Maximum design mass	10,000 kg	Program target; requires mass closure
Rotor count	7	Distributed lift architecture
Rotor diameter	5.6 m	Preliminary geometric target
Overall envelope	17 m × 17 m × 4 m	Preliminary packaging envelope
Fuel capacity	1,600 L	Nominal tank capacity; density varies with fuel
Primary capsule	Fixed / integrated	Mechanical isolation and energy attenuation
Future capsule	Separable / hover-capable	Derivative, not Document 00 baseline
Architecture	Distributed propulsion	Power distribution and control are central design issues
Design philosophy	Multi-layer safety	Independent layers, graceful degradation

3. What Changed from the 200-Ton Program

The earlier 200-ton concept and Documents 00–05 associated with that architecture are explicitly cancelled for this program baseline. Their numerical design assumptions must not be copied into the 10-ton aircraft without re-derivation.

The new platform is not simply a scaled-down version. It is a different product architecture: fewer structural loads, shorter development loops, smaller propulsion units, simpler ground handling, and a demonstrator-sized path to validation.

Area	Earlier direction	10T direction
Mass	200 t class	10 t class
Safety capsule	Complex heavy architecture	Integrated fixed capsule first
Propulsion	Heavy distributed system	Distributed electric or mechanical hybrid architecture
Prototype objective	Full-scale heavy platform	Technology demonstrator / flagship prototype
Commercial story	Long-horizon aircraft program	Visible, testable product milestone
Development risk	Very high	Lower, but still aerospace-critical

This change improves program tractability, but it does not make certification or flight testing easy. A 10-ton rotorcraft remains a major aerospace system.

4. Architecture Overview

The proposed aircraft is a distributed-lift platform in which seven rotors share the lift task. The protected capsule occupies the central survivable volume. Propulsion, fuel, electrical conversion, thermal systems and control electronics are placed around the capsule so that no single peripheral subsystem becomes the sole occupant-survival path.

A preferred conceptual architecture is four prime movers feeding an electrical distribution system, with power electronics allocating energy to seven rotor drive units. This is a benchmark architecture, not a finalized engine installation.

High-level flow: Prime movers → generators → DC/AC power conversion → independent rotor drive channels → seven rotors. Safety systems supervise every layer and default toward controlled degradation rather than a single catastrophic failure mode.

Subsystem	Preferred concept	Design objective
Prime movers	4 turboshaft-class units	Redundancy and serviceability
Generation	Independent generator channels	Prevent single electrical fault from disabling all lift
Rotor drives	7 independent channels	Distributed thrust / fault isolation
Flight control	Triplex/segregated digital control concept	Fault containment
Capsule	Fixed integrated survival cell	Occupant protection
Energy attenuation	Mechanical + structural layers	Manage crash energy
Fire protection	Zone-based detection/isolation	Prevent propagation

5. Preliminary Mass Closure

A 1,600 L fuel load is a major mass item. At an illustrative fuel density of 0.80 kg/L, the fuel itself is approximately 1,280 kg. Actual aircraft fuel mass must use the specified fuel grade, temperature and tank condition.

The preliminary target below is deliberately treated as a mass budget, not a claim that the aircraft can already meet it. The 10,000 kg ceiling creates a hard design constraint: capsule, rotor system, powertrain and structure must be aggressively weight-controlled.

Mass item	Target kg	Comment
Fixed safety capsule + restraints	900	Protected occupant cell; preliminary
Primary airframe / load paths	1,100	Frames, skins, mounts; preliminary
Seven rotor systems	1,200	Blades, hubs, bearings, drive interfaces
Prime movers + installed accessories	900	Benchmark for four-unit architecture
Generators + power electronics	450	Preliminary distributed system allowance
Fuel system + tank hardware + fuel	1,400	Includes ~1,280 kg illustrative fuel mass
Landing gear / flotation / survival	300	Preliminary
Avionics / sensors / control	250	Preliminary
Fire protection / emergency systems	180	Preliminary
Thermal management / exhaust	180	Preliminary
Cabin / utilities / interior	200	Minimal demonstrator fit
Water-survival provisions	100	Concept allowance
Program mass contingency	650	Must be consumed only after design maturation
TOTAL TARGET	7,810	Leaves ~2,190 kg below 10,000 kg ceiling

Mass closure warning: the remaining 2,190 kg is not automatically payload. It is reserve for growth, certification hardware, test instrumentation, integration changes and uncertainty. A credible aircraft configuration must eventually close the mass statement with defined empty mass, operating equipment, crew, fuel state and payload.

6. Fuel and Mission Mass Logic

The 1,600 L tank should be treated as a maximum nominal capacity rather than a promise that every mission will depart with full tanks. Carrying unused fuel is structurally and energetically expensive.

For a 10,000 kg gross-mass target, a full 1,600 L load at 0.80 kg/L consumes approximately 1,280 kg. This leaves 8,720 kg for everything else at maximum gross mass. That arithmetic should govern future design trades.

Fuel state	Illustrative fuel mass	Program implication
1,600 L	~1,280 kg	Full-tank reference
1,200 L	~960 kg	Potential demonstrator mission state
900 L	~720 kg	Lower-mass test state
700 L	~560 kg	Shorter test / handling state

Recommended development approach: certify the tank volume and fuel system architecture independently, but fly early tests with reduced fuel mass wherever the test objective allows it.

7. Propulsion Benchmark and Engine Selection

For the present concept, a four-prime-mover distributed-electric architecture is the strongest starting point because it allows the seven rotor channels to remain modular while retaining engine-level redundancy. A direct mechanical seven-way gearbox would be lighter in some regimes but introduces difficult fault-containment and mechanical synchronization problems.

The benchmark engine selected for preliminary system studies is the Pratt & Whitney Canada PT6C-67A class. P&WC; publishes approximately 1,940 mechanical shaft horsepower and 1,166 kW takeoff power for the PT6C-67A, with an approximate dry mass of 207 kg in the EASA type-certificate data. These are published engine figures, not installed-aircraft performance.

Candidate	Published power class	Why it matters
PT6C-67A	~1,166 kW takeoff / ~1,024 kW max continuous	Strong benchmark; four units create substantial redundancy margin
PT6C-67E	~969 kW takeoff / ~900 kW max continuous	Lower-power alternative; mature helicopter application
Rolls-Royce M250 Series IV	420–715 SHP family range	Compact, proven, but four units may under-power the present target unless many units are used
Safran Aneto family	Up to 2,500 SHP class in cited Aneto-1K	Powerful, but larger than necessary for a first 10T demonstrator

Preliminary choice: 4 × PT6C-67A-class prime movers as the analytical benchmark. This is not an endorsement of procurement or a final engine selection. Engine availability, certification pathway, integration rights, generator compatibility, gearbox requirements, maintenance support, export controls and total installed mass must be resolved with the OEM.

8. Why Four Prime Movers + Seven Rotors

Four prime movers are a practical compromise between redundancy and system complexity. The design does not depend on one engine. Power is pooled or selectively routed so that a single engine loss can become a degraded-power event rather than an immediate total-power event.

Seven rotors create an odd-numbered distributed-lift field with a central rotor line and six surrounding stations in the conceptual layout. The exact geometry, rotor overlap, inflow interaction and control allocation require CFD, rotorcraft analysis and flight-test validation.

Failure	Desired response	Design intent
One prime mover lost	Reallocate available power	Controlled degradation
One generator channel lost	Isolate channel; preserve remaining bus segments	Fault containment
One rotor drive lost	Redistribute lift/control	Distributed redundancy
Single inverter fault	Isolate affected rotor channel	Prevent bus collapse
Sensor disagreement	Voting / fault isolation	Avoid single-sensor authority
Local fire detected	Fuel/electrical isolation + suppression	Prevent propagation

A critical design rule is that redundancy must be physically independent. Seven identical channels sharing one vulnerable bus, shaft, cooling loop or controller does not equal seven independent safety channels.

9. Ten-Layer Safety Architecture

The ten-layer architecture is retained as the core identity of the aircraft. The layers are intentionally overlapping. Each layer has a different failure mechanism and should be independently testable.

Layer	Function	Concept
1	Occupant survival capsule	Protected central volume with dedicated load paths.
2	Multi-stage energy absorption	Mechanical dampers, crushable structures and controlled load paths.
3	Capsule-to-airframe isolation	Isolation interfaces reduce transmission of impact and vibration.
4	Distributed propulsion	Seven rotor channels reduce dependence on one lift device.
5	Prime-mover redundancy	Multiple independent power sources.
6	Fire detection / isolation	Zone-based detection, fuel shutoff and electrical isolation.
7	Control redundancy	Independent sensing, computation and actuator paths.
8	Emergency descent / recovery	Controlled degraded flight and emergency recovery functions.
9	Water survival	Buoyancy, flotation and occupant egress provisions.
10	Structural containment / survivability	Rotor, fuel and energy hazards kept away from the occupant volume.

The phrase “10-layer safety” is a design philosophy, not a safety certification claim. Each layer must ultimately be backed by requirements, analyses, tests, failure-mode evidence and certification authority acceptance.

10. Fixed Safety Capsule: Primary Architecture

The flagship demonstrator uses a non-separable occupant capsule. The capsule is therefore not required to perform an independent flight, hover or landing sequence. This removes a large class of propulsion, separation, control and synchronization hardware from the first aircraft.

The capsule is envisioned as a structurally protected volume connected to the airframe through controlled mechanical interfaces. The design objective is to reduce peak occupant loads and maintain a survivable volume after a severe landing or crash event.

The capsule should not be described as “crash-proof.” The engineering target is a defined survivability envelope: preserve occupant space, manage deceleration, prevent secondary hazards, and maintain viable egress paths within the tested envelope.

Feature	Baseline concept
Capsule attachment	Multiple distributed mechanical interfaces
Primary attenuation	High-stroke mechanical dampers
Secondary attenuation	Controlled deformation / crush structures
Seat interface	Energy-attenuating occupant restraint system
Fuel separation	Fuel volume outside the primary occupant cell
Fire barrier	Thermal/fire-resistant separation between hazard zones and capsule
Egress	At least two conceptually independent exit routes where geometry permits

11. Mechanical Energy Absorption Strategy

The supplied concept includes thick and thin fiber/composite structural elements, multiple mechanical dampers and dedicated capsule support points. The important systems-engineering principle is to create a sequence of energy-management events rather than asking one component to absorb the entire crash energy.

A conceptual sequence is: rotor/landing-system energy dissipation → airframe deformation → capsule-interface stroke → capsule structural load distribution → seat/restraint energy management. The actual sequence depends on impact attitude and must be validated experimentally.

Stage	Mechanism	Desired outcome
A	Landing/outer structure response	Remove energy before capsule loading
B	Primary airframe deformation	Control load path and avoid brittle collapse
C	Capsule support stroke	Lengthen deceleration event
D	Capsule shell/load spread	Preserve survivable volume
E	Seat/restraint attenuation	Reduce occupant acceleration exposure

A real crashworthiness program must define impact cases, pulse shapes, structural limits, occupant injury criteria and test instrumentation. Those values are intentionally not asserted here because they require a qualified crashworthiness analysis and regulatory basis.

12. Rotor and Propulsion Safety

The seven rotors are major energy sources. Rotor retention, blade failure, hub failure, overspeed, drivetrain failure and foreign-object events must be treated as primary hazards.

The preferred safety philosophy is containment and isolation. The design should avoid routing critical occupant-survival structures through rotor hazard zones wherever possible and should separate rotor drive channels so one failure cannot mechanically cascade into neighboring channels.

Hazard	Conceptual mitigation
Blade failure	Rotor-zone containment philosophy; validated blade/hub design
Overspeed	Independent monitoring and shutdown / power limiting
Drive seizure	Channel isolation and structural torque-path control
Inverter fault	Fast electrical isolation
Generator failure	Power reallocation
Rotor control failure	Independent control-channel isolation

Rotor safety is one of the highest-priority engineering work packages. The 5.6 m rotor diameter is a geometry input, not proof that the rotor system is safe.

13. Fire, Fuel and Thermal Safety

The 1,600 L fuel system is one of the largest stored-energy hazards. The safety architecture therefore separates fuel zones from the capsule and uses compartmentalization, detection, isolation and controlled venting as system-level objectives.

Thermal management must consider engine exhaust, generators, inverters, wiring, bearings, hydraulic or electric actuators, and hot surfaces. Fire protection cannot be limited to an engine nacelle because energy can propagate through wiring, fuel plumbing and structural cavities.

Zone	Protection concept
Fuel tanks	Segregated tank cells; protected lines; leak detection concept
Engine bays	Fire detection, fuel isolation and suppression concept
Power electronics	Thermal monitoring, current limiting, isolation
Capsule boundary	Fire/thermal barrier and controlled penetrations
Exhaust	Thermal shielding and hot-zone separation
Emergency shutdown	Centralized and local isolation logic

14. Flight-Control and Fault-Reconfiguration Philosophy

The aircraft should be designed around fault containment rather than perfect-component assumptions. The flight-control system should continuously determine which propulsion and sensing channels are healthy, degraded or unavailable.

The seven-rotor configuration gives the control allocator multiple degrees of freedom, but this is only useful if the actuator, sensor, computation and power paths are sufficiently independent.

Conceptual control hierarchy: mission control → vehicle state estimation → fault manager → control allocator → rotor-channel commands → local actuator control. A fault manager should be able to quarantine a failed channel without allowing the fault to contaminate the rest of the control system.

Control function	Desired architecture
State estimation	Multiple independent sensor sources
Control computation	Segregated processing channels
Actuation	Independent rotor channels
Power	Separated distribution paths
Fault management	Explicit health-state machine
Emergency mode	Simplified degraded-control law

15. Water Survival and Emergency Landing

The supplied concept includes water-survival capability as a dedicated safety layer. The fixed capsule architecture is advantageous here because the occupant cell remains mechanically connected to the aircraft and can be designed around flotation and controlled egress rather than separation dynamics.

Water survival should address buoyancy after structural damage, rollover orientation, ingress, occupant restraint release, emergency lighting, communication and access to exits. These are design requirements, not completed features.

A useful test philosophy is to treat water survival as a separate subsystem validation program. The aircraft should not rely on one flotation device or one hull region to maintain survivability after a crash.

16. Power Architecture and Efficiency Trade

For the seven-rotor concept, the most attractive preliminary architecture is a distributed electrical power network supplied by multiple turboshaft-driven generators. This allows the prime movers to operate in a narrower efficient operating band while rotor speed and power demand are managed independently.

The trade is not simply “turboshaft versus turbogenerator.” A turboshaft mechanically driving a rotor can be highly efficient, while a turboshaft-generator-electric-motor chain adds conversion losses but can greatly simplify power distribution and fault isolation.

Architecture	Strength	Main penalty
Direct mechanical	Fewer conversion stages	Complex mechanical distribution / fault propagation
Central gearbox + rotors	Compact power path	Large single-point mechanical dependency
4 turboshaft + electric distribution	Modularity + redundancy	Generator/inverter mass and conversion losses
Hybrid electric	Flexible energy management	Battery mass and thermal complexity

Preliminary recommendation: continue with the four-engine electrical architecture for the prototype studies because it aligns best with the safety philosophy and seven independent rotor channels. Optimize total system efficiency after a realistic installed-mass model exists.

17. Engine Benchmark: What the Published Data Says

Pratt & Whitney Canada states that the PT6C family uses a two-shaft layout with a multi-stage compressor, a compressor turbine and an independent power turbine. The company lists the PT6C-67A at approximately 2,000 shaft horsepower thermodynamic class and 1,940 shaft horsepower mechanical class.

EASA type-certificate data for the PT6C-67 series lists the PT6C-67A at approximately 1,166 kW takeoff power and 1,024 kW maximum continuous power, with an approximate dry specification weight of 207.3 kg. These are engine-level values and do not include the complete aircraft installation.

Conceptual calculation	Result
4 × 1,024 kW max continuous	≈ 4,096 kW engine-level
4 × 1,166 kW takeoff	≈ 4,664 kW engine-level
2,800–3,300 kW aircraft target	Within benchmark engine-level envelope
Installed system	Lower after accessories, generators, conversion and thermal constraints

This is why the PT6C-67A class is a useful benchmark. It does not mean the aircraft automatically produces 4 MW at the rotors. Generator efficiency, gearbox losses, inverter losses, motor losses, thermal limits, altitude and installation effects must be modeled.

18. Comparative Engine Candidates

The M250 family is exceptionally attractive for compactness and service history, but its published 420–715 SHP range means a 10-ton, seven-rotor aircraft would require a larger number of prime movers or a lower system-power target. That increases plumbing, controls and maintenance burden.

The Safran Aneto family sits in a much higher power class. Safran cites a 2,500 SHP Aneto-1K selected for the AW189K. It offers ample power but is likely oversized for the first 10-ton demonstrator unless the architecture evolves toward a much higher gross mass or a low-engine-count mechanical system.

Candidate	Fit to 10T demonstrator	Preliminary status
PT6C-67A class	Strong	Primary benchmark
PT6C-67E class	Moderate	Alternative / lower-power study
M250 Series IV	Low-to-moderate	Compact alternative, likely more units
Safran Aneto-1K class	Low for first prototype	High-power future option

The next engineering gate is not “pick an engine from a brochure.” It is “freeze a required shaft/electrical power curve, then negotiate an engine/generator package against it.”

19. Preliminary Performance Logic

The previous project material included indicative values such as roughly 700–800 kg installed-power-class references per unit, 2.8–3.3 MW total installed power, and 60–70 minute hover-class durations in some configurations. Those earlier values are not carried forward as validated performance numbers.

For Document 00, performance should be derived from the new 10-ton mass, actual rotor disk area, blade loading, atmospheric condition, drivetrain efficiency, engine power lapse, and mission reserve policy.

Quantity	What must be solved
Hover power	Momentum/blade-element/CFD model plus installation effects
Climb	Excess power at defined mass and atmosphere
Cruise	Drag model + rotor/propulsive efficiency
Endurance	Fuel flow versus power schedule
Range	Cruise fuel flow + reserves + routing
OEI / degraded case	Available power after one channel failure

The correct answer to “best efficiency” is therefore a system optimum, not a single engine brand. The rotor disk loading and installed propulsion efficiency may dominate the aircraft’s total energy use.

20. Development and Verification Roadmap

The 10-ton concept should be developed through staged evidence. The first objective is to retire the most dangerous unknowns before committing to a full-scale flying prototype.

Stage	Evidence gate
1. Requirements freeze	Mass, rotor geometry, power, capsule volume, safety requirements
2. Digital architecture	Mass properties, power flow, failure modes, thermal model
3. Subscale rotor test	Rotor efficiency, vibration, control authority
4. Capsule drop / impact tests	Energy absorption and occupant-cell survivability
5. Propulsion bench	Engine-generator integration and fault isolation
6. Full-scale static ground rig	Power, thermal, vibration, emergency shutdown
7. Tethered / constrained test	Low-risk control and distributed-lift validation
8. Incremental flight envelope	Progressive expansion under qualified test program

No flight test should be inferred from the existence of this document. A qualified test organization must define the actual test program, instrumentation, operating limits and go/no-go criteria.

21. Preliminary Risk Register

The following risks are deliberately prominent because they can invalidate the concept if ignored. They should become a living program register.

Risk	Severity	Mitigation direction
Mass growth	High	Configuration control + component-level mass budgets
Rotor structural failure	Critical	Dedicated rotor/hub test and containment analysis
Single-point electrical fault	High	Segmented buses and independent channels
Capsule attachment overload	Critical	Crashworthiness analysis + full-scale tests
Thermal runaway / fire propagation	High	Zone isolation + detection + suppression
Engine integration mismatch	High	OEM engagement + generator matching
Control-channel common cause	High	Physical/software segregation
Water rollover / egress	High	Dedicated water-survival test campaign
Certification burden	High	Early regulatory engagement

22. Manufacturing Philosophy

The 10-ton version should be designed for manufacturing simplicity from the beginning. The objective is not merely to make a lighter aircraft, but to reduce the number of precision interfaces and one-off parts.

A modular rotor pod, repeatable capsule interface, standardized electrical channel and replaceable power module are preferred over deeply integrated assemblies. This supports maintenance, testing and iterative prototyping.

Module	Preferred strategy
Rotor unit	Replaceable module with standardized interfaces
Power unit	Engine + generator module
Inverter	Line-replaceable electrical module
Capsule	Independent structural subassembly
Landing/flotation	Replaceable energy-management modules
Avionics	Separated compute/sensor modules

A prototype that can be disassembled into understandable modules is strategically valuable because each module can become a test article, supplier package or future IP family.

23. Commercial and IP Positioning

The commercial proposition is strongest when the aircraft is presented as a demonstrable safety-and-propulsion architecture rather than as a promise of a finished certified aircraft.

The fixed capsule is the lead product because it removes the complexity of capsule separation while keeping the core safety story visible. The separable capsule can remain a derivative technology path after the fixed-capsule demonstrator proves the fundamental architecture.

Near-term asset	Purpose
10T fixed-capsule demonstrator	Visible proof of architecture
Safety architecture package	System-level IP / engineering evidence
Distributed propulsion module	Reusable platform technology
Capsule attenuation system	Potential standalone technology family
Future separable capsule	Higher-complexity derivative

The program should avoid presenting speculative numerical performance as contractual capability. The strongest investor or industrial-partner story is a credible sequence of technical proof points.

24. Requirements to Freeze Before Document 01

Document 00 is intentionally a foundation. Before detailed subsystem documents are generated, the following values should be frozen or explicitly marked as open:

- Exact MTOM and whether 10,000 kg is a hard limit or a nominal target.
- Occupant count and mission payload definition.
- Rotor geometry beyond diameter: blade count, chord distribution, tip speed and hub architecture.
- Prime-mover architecture: mechanical shaft versus generator output.
- Required continuous, takeoff and degraded-case power.
- Fuel type and reference density.
- Capsule dimensions and occupant loading assumptions.
- Landing impact cases and crashworthiness test philosophy.
- Water-survival requirement and flotation architecture.
- Certification category and intended operating environment.
- Thermal limits and allowable power-electronics operating temperatures.
- Maintenance and module-replacement philosophy.

25. Preliminary System Requirement Set

The following are candidate requirements for further engineering review. They are intentionally written as design targets rather than compliance statements.

ID	Candidate requirement
SYS-001	The aircraft shall target a maximum design mass of 10,000 kg.
SYS-002	The aircraft shall use seven distributed lift rotors in the baseline configuration.
SYS-003	Baseline rotor diameter shall be approximately 5.6 m pending aerodynamic optimization.
SYS-004	Nominal fuel tank capacity shall be approximately 1,600 L, subject to mass and structural closure.
SYS-005	The baseline occupant capsule shall remain mechanically integrated with the aircraft.
SYS-006	The safety architecture shall include ten distinct, reviewable protection layers.
SYS-007	No single rotor-channel failure shall automatically imply total loss of all lift authority.
SYS-008	Fuel, fire and electrical hazards shall be segregated from the primary occupant volume.
SYS-009	The design shall support controlled degraded operation after selected subsystem failures.
SYS-010	All performance values shall remain provisional until validated by analysis and testing.

26. Source Notes and Engineering Boundaries

Project-source basis: the supplied screenshots and user-provided revision defining the 10-ton program, seven 5.6 m rotors, 1,600 L fuel capacity, 17 m × 17 m × 4 m envelope, fixed integrated capsule, layered mechanical energy absorption, distributed propulsion and the ten-layer safety philosophy.

External benchmark sources used for engine context:

• Pratt & Whitney Canada, PT6C Engine product page: PT6C family architecture and published power classes.

• EASA, Type Certificate Data Sheet IM.E.022, PT6C-67 series: engine dimensions, dry weights and certified power ratings.

• Rolls-Royce, M250 Turboshaft: published M250 family architecture, service history and power range.

• Safran, Aneto engine family announcement: Aneto-1K 2,500 SHP class and super-medium/heavy helicopter positioning.

Engineering boundary: external engine data is used only as a benchmark. It does not establish suitability, availability, certification, installation compatibility, fuel consumption, generator integration, or procurement rights for this aircraft.

27. Founder Decision Summary

The 10-ton fixed-capsule aircraft is the recommended flagship configuration for the next development phase because it retains the program's strongest differentiator while removing the hardest feature from the first build: a fully separable, independently hovering safety capsule.

The strongest preliminary propulsion architecture is four turboshaft-class prime movers feeding a segmented electrical distribution system and seven independently controlled rotor channels. The PT6C-67A class is the current analytical benchmark because its published power class fits the 2.8–3.3 MW system-level direction with substantial engine-level margin, while maintaining a manageable engine count.

The central engineering challenge is mass closure. A full 1,600 L fuel load can represent roughly 1.28 tonnes at an illustrative 0.80 kg/L density. The design therefore has to protect the capsule, rotor system and powertrain without allowing the 10-ton target to dissolve under subsystem growth.

PROGRAM NORTH STAR

Build the smallest aircraft that can visibly prove the biggest idea: a distributed-lift vehicle whose occupant-protection architecture is treated as a primary system, with multiple independent layers between failure and human harm.

28. Document Control / Next Revision

Document 00 establishes the conceptual baseline only. The next revision should convert the program narrative into a traceable requirements set and a controlled mass/power model.

Field	Value
Document number	00
Program	10T Safety-Capsule Distributed-Lift Aircraft
Founder	Amirbahador Atai
Baseline	Fixed integrated safety capsule
Rotor count	7
Rotor diameter	5.6 m target
Fuel capacity	1,600 L nominal target
Envelope	17 m × 17 m × 4 m target
Primary engine benchmark	4 × PT6C-67A-class turboshaft/generator concept
Status	Conceptual / non-certified
Next document gate	Requirements + mass/power closure

END OF DOCUMENT 00

Founder Amirbahador Atai